# CATH PLAN · Case 2

74M, 78 kg · scheduled 09:30 · PCI to mid-LAD (NSTEMI) · pathway: known intervention

**Synthetic teaching content. Not clinical advice. Not a medical device. Approved by the chair on the Switch card; still a planning draft, not an order.**

**Thresholds: PROPOSED values, not yet verified by the chair. Template logic (the Heart Team factor rules, stage assignment, Labs' eGFR cutoff) is also unreviewed.**

| 7 | 7.1 | 3 / 4 | 0 |
|---|---|---|---|
| flags from 4 agents | contrast ratio (proposed cap 3.7) | angio findings flagged (calcium, flow, bifurcation, multivessel) | devices out of stock |

Contrast ratio = planned contrast (mL) ÷ eGFR. The published ratios use creatinine clearance, not eGFR (docs/EVIDENCE.md E1, E2): risk rose from 2 (Gurm 2011) and 3.7 was a discriminating cutoff (Laskey 2007). No guideline sets a limit. The chair confirms the basis. The packet flags this eGFR as unreliable while creatinine is changing.

## Lesion map (Imaging)

| Lesion | Vessel | Stenosis | TIMI | Calcium | Flag |
|---|---|---|---|---|---|
| L1 | mid-LAD | 90% | TIMI 2 | calcium moderate | none |
| L2 | mid-LCx | 70% | TIMI 3 | calcium none | none |

Triggers: calcium moderate or severe; TIMI flow below 3; more than one lesion (multivessel). · Cath report: indication NSTEMI; access radial; hemodynamics not recorded

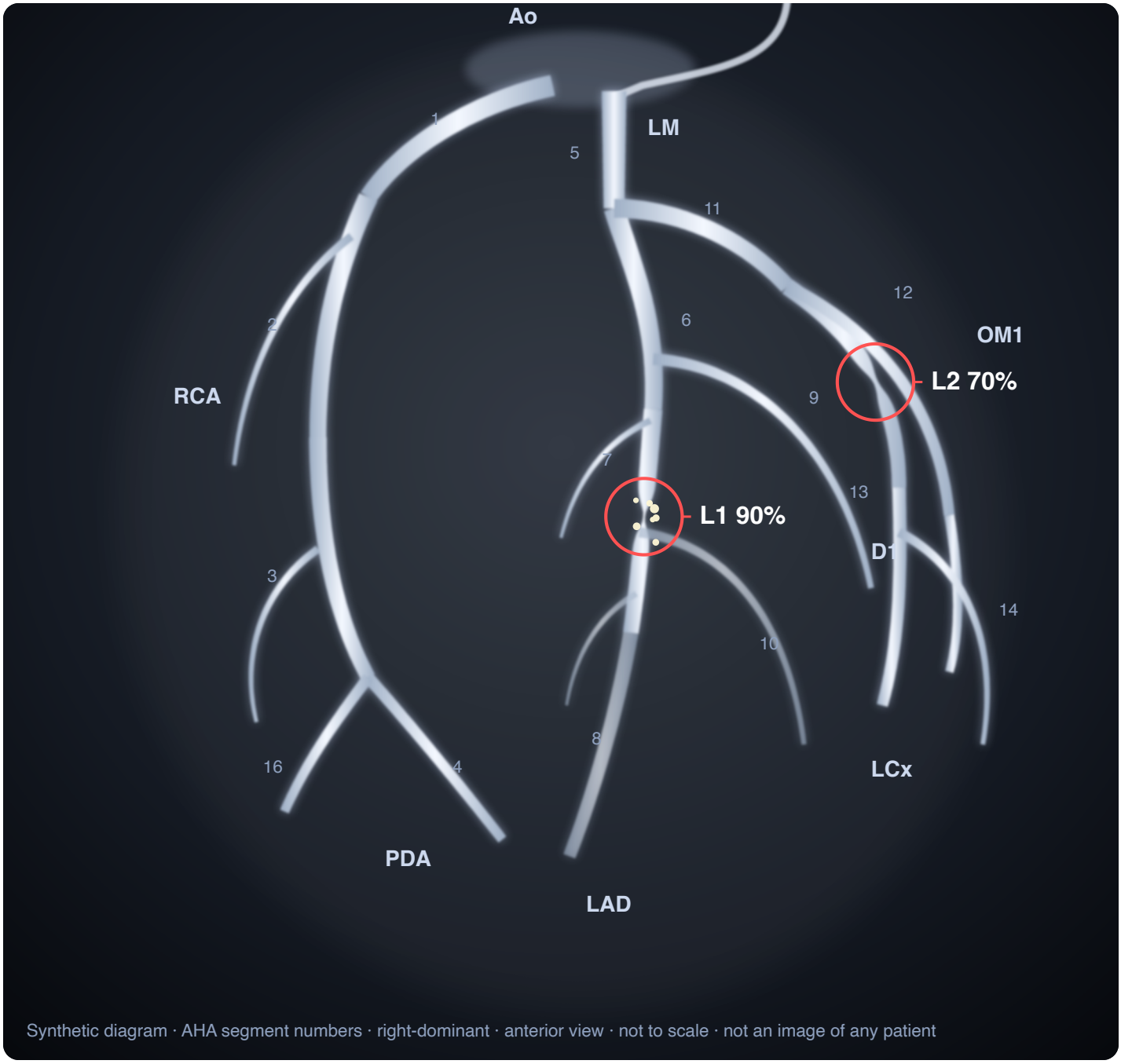
## Rules that fired

- Contrast rule exceeded: ratio 7.1 (170 mL ÷ eGFR 24) is above the proposed cap 3.7 (Laskey 2007 cutoff, E2) and above 3 (Gurm 2011, E1; both use creatinine clearance, not eGFR). Chair to decide whether to stage.

## If the chair stages (illustration from the packet, not a recommendation)

| Stage | Lesions | Devices |
|---|---|---|
| **Stage 1** | L1 mid-LAD 90% (the procedure named in the packet) | 7F guide, NC balloon 3.5 mm, DES 3.5x28 mm, standard PCI tray |
| **Stage 2** | L2 mid-LCx 70% | **not listed in the packet: Equipment to confirm** |

## Angiographic findings (synthetic)




**L1 · mid-LAD** (AHA segment 7)
90% stenosis · TIMI 2 (slow filling: the distal vessel is drawn fainter) · calcium moderate · flag none

**L2 · mid-LCx** (AHA segment 13)
70% stenosis · TIMI 3 (normal filling) · calcium none · flag none

Drawn from the angio lines in the packet by tools/angio.py: right-dominant tree, anterior view, AHA segment numbers, lumen narrowed to the stated percentage, calcium shown as bright specks, distal filling dimmed when TIMI flow is below 3. A teaching diagram, not an image of any patient and not to scale.

## Equipment manifest

| Item (from the packet) | Stock |
|---|---|
| 7F guide | 3 on hand |
| NC balloon 3.5 mm | 3 on hand |
| DES 3.5x28 mm | 2 on hand |
| standard PCI tray | 3 on hand |

## Prep check (devices against findings)

| | |
|---|---|
| **Scope** | Stage the devices named in the packet |
| **Stock** | all named devices in stock |

| On the shelf, by class (from the STOCK line) | |
|---|---|
| Guide catheters | 7F guide 3, 6F guide 3 |
| Diagnostic catheters | 5F diagnostic catheters 6 |
| Wires | support wire 3 |
| Balloons | NC balloon 3.5 mm 3, NC balloon 3.0 mm 3 |
| Stents (DES) | DES 4.0x24 mm 2, DES 3.5x28 mm 2, DES 3.0x18 mm 2 |
| Intravascular imaging | IVUS catheter 0 (resupply ETA 13:00) |
| Support devices | Impella CP 2 |

**Questions for the chair** (the plan asks, it does not choose a device):

- 2 lesions on angiography and 1 stent named in the packet: chair to confirm what to stage.
- Calcium moderate on L1 mid-LAD (angiography): the packet names no calcium-modification device or intravascular imaging catheter. Chair to decide whether any is needed.
- No wire is named in the packet's device lines: chair to confirm.
- Calcium or left main on angiography, and IVUS catheter is 0 on hand with resupply ETA 13:00, after the scheduled 09:30 start: chair to decide how to proceed.

## Flags from the agents

**Imaging**
- calcium moderate · L1 mid-LAD 90%
- TIMI below 3 · L1 mid-LAD 90% TIMI 2
- multivessel · L1 mid-LAD 90% · L2 mid-LCx 70%

**Labs**
- Cr 2.7 (2.2 yesterday) · Est. contrast 170 mL
- eGFR 24 · Est. contrast 170 mL

**Meds**
- heparin infusion running · radial access
- metformin on the list · Contrast given at the diagnostic angiogram yesterday

**Equipment** ✅ nothing flagged

Relay: Imaging → Equipment, Labs → Meds.

## Summary and approvals

| | |
|---|---|
| **Ruling recorded** | none yet |

| | |
|---|---|
| **Status** | ✔ Chair ruling recorded: approved on the Switch card, 2026-11-04 20:11. |
| **Flags** | Imaging 3 · Labs 2 · Meds 2 · Equipment 0 |
| **Rules that fired** | Contrast rule exceeded: ratio 7.1 (170 mL ÷ eGFR 24) is above the proposed cap 3.7 (Laskey 2007 cutoff, E2) and above 3 (Gurm 2011, E1; both use creatinine clearance, not eGFR). Chair to decide whether to stage. |

**For the chair to decide / check** (the agents only flag):

- Thresholds in thresholds.json are proposed: the chair verifies them before this plan is used.
- Template logic (the Heart Team factor rules, the stage assignment, Labs' eGFR cutoff) is unreviewed.
- Stage 2 devices are not listed in the packet: Equipment to confirm.

**Approvals**

| Time | Request | Decision | By |
|---|---|---|---|
| 20:11 | this plan | Allow once | the chair, on the Switch card |

**Status**
Approved by the chair on the Switch card.

APPROVED BY CHAIR

Sources: packet (day.md) · briefs from Imaging, Labs, Meds, Equipment · thresholds.json · evidence table docs/EVIDENCE.md (E1 to E15, secondary summaries, chair verifies). Generated 2026-11-04 20:11 by plan.py (template, no model).